

Theory and Modelling of Large-Scale Modes of Variability in the Atmosphere

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Balance

- ⑥ Bringing balance to a session by *not* talking about construction of Balance Models.
- ⑥ Unimportance of gravity waves in the theory and modelling to be presented.

Some Problems

1. What is the nature and cause of low-frequency variability ($\sim 10 - 30$ days) in the atmosphere?
 - ⑥ Its temporal structure: time-scales longer than that of baroclinic instability, or of propagating stationary waves.
 - ⑥ Its spatial structure: preferred patterns or modes of variability, such as so-called annular modes, the NAO, etc.
2. What are the underlying dynamics?
 - ⑥ Influence of underlying boundary conditions? Sea-surface temperature anomalies?
 - ⑥ Barotropic and equivalent barotropic dynamics vs. baroclinic effects.
 - ⑥ Wave-meanflow interaction and geostrophic turbulence.

Previous work: Wallace (QJ 2001); DeWeaver and Nigam (JAS, 2000); Lorenz and Hartmann (JAS 2002); Feldstein, Lee (various); others.

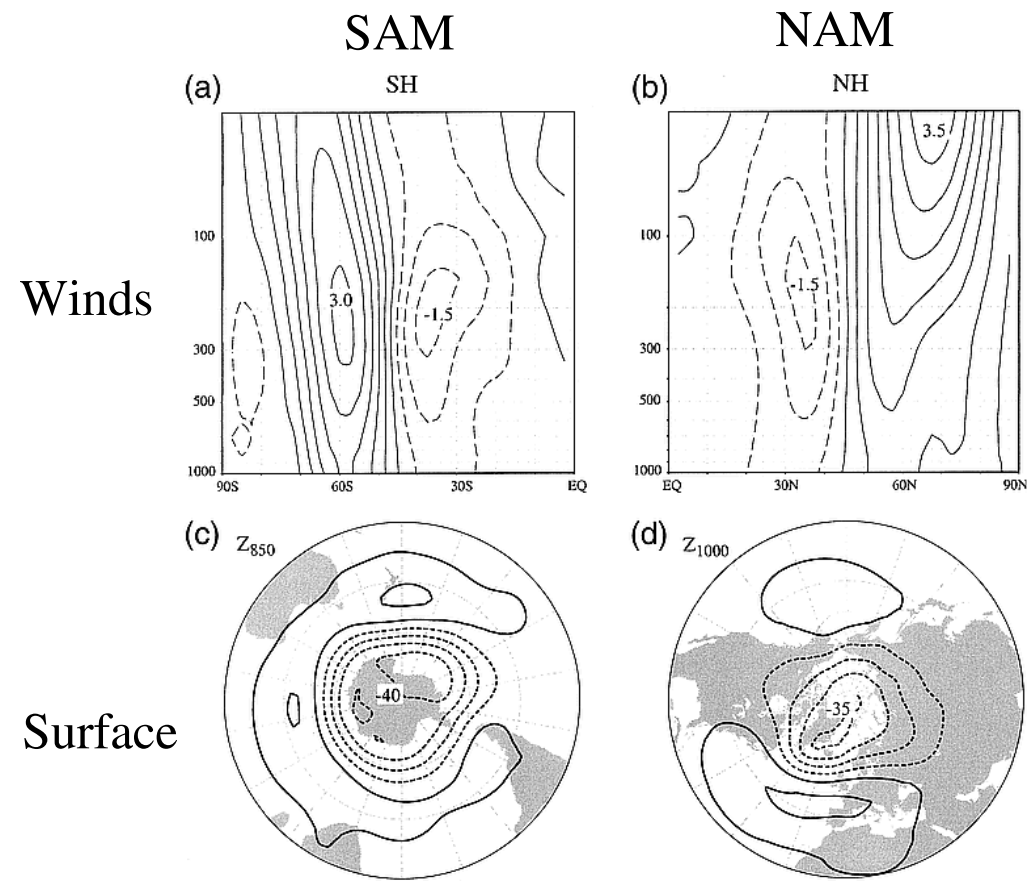
Observations

There are distinct patterns of variability in the atmosphere, for example:

- (i) The North Atlantic Oscillation (NAO); the Pacific Decadal Oscillation; the Southern Hemisphere Annular Mode (SHAM).
- (ii) Often identified by EOF analysis, and by teleconnections (analysis of correlations).
- (iii) A particular current controversy revolves around whether some of the modes are just regional manifestations of modes of hemispheric extent. *Do so-called annular modes exist?* (Wallace, 2000; Ambaum *et al.*, 2001)

Hemispheric Modes of variability

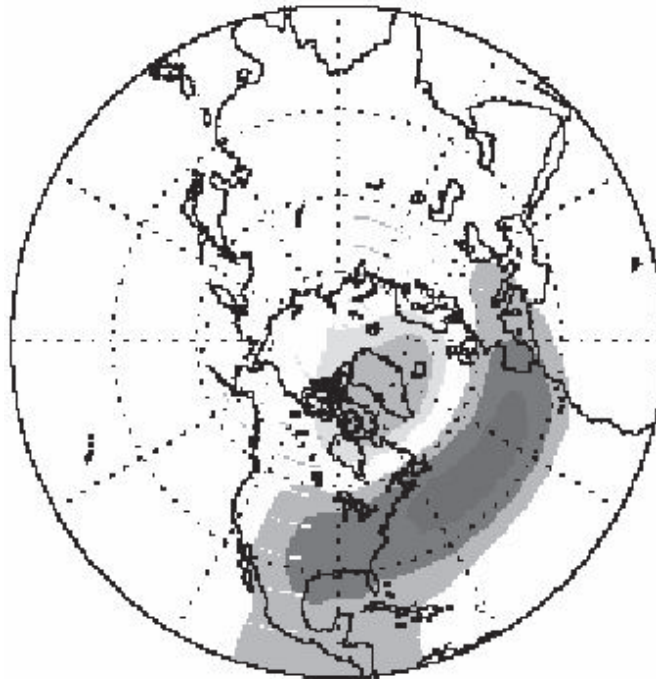
Defined as Leading EOF of sea level pressure



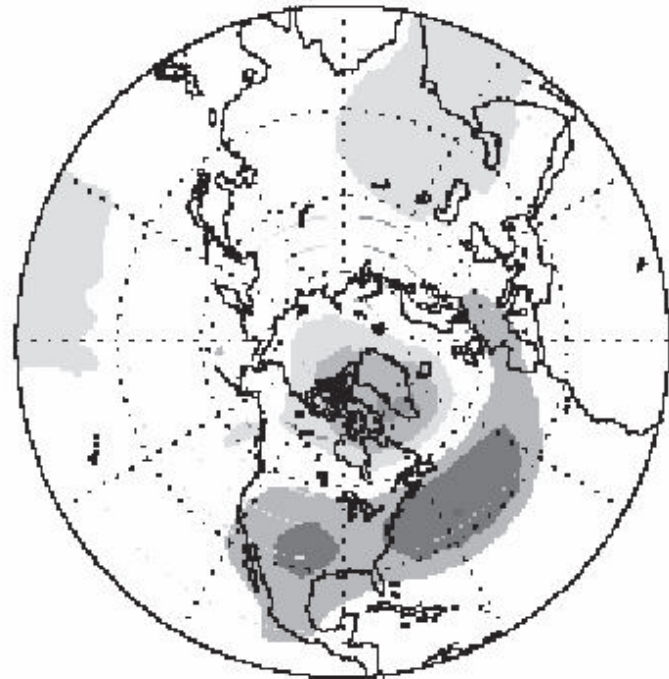
Thompson and
Wallace (2000)

NORTH ATLANTIC OSCILLATION (NAO)

January

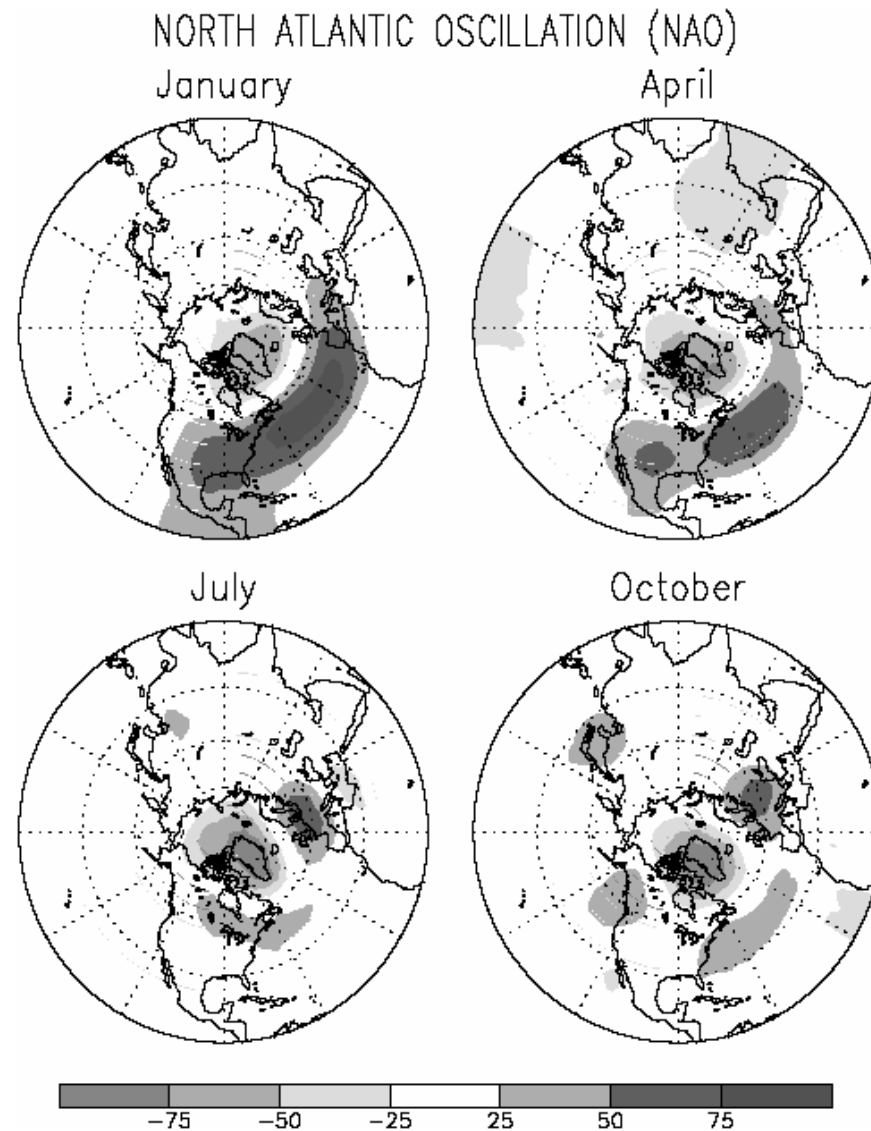


April



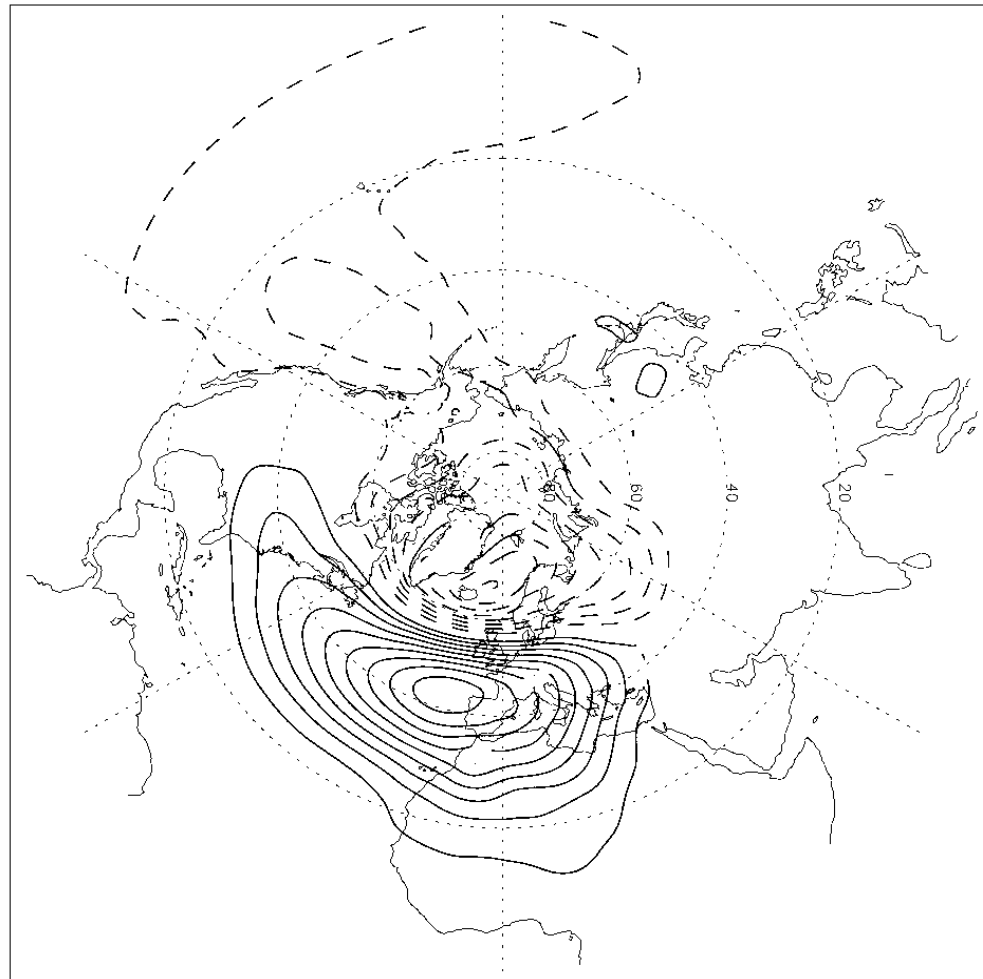
Spatial pattern of the NAO, following Barnston & Livezey 1987.

The panels show the NAO as determined from a rotated EOF analysis of the 700 mb anomaly height field, for January and April from 1964–1994.



Spatial pattern of the NAO, following Barnston & Livezey, 1987.

A rotated EOF analysis of the 700 mb anomaly height field.



EOF (from Ambaum et al 2001) showing the NAO

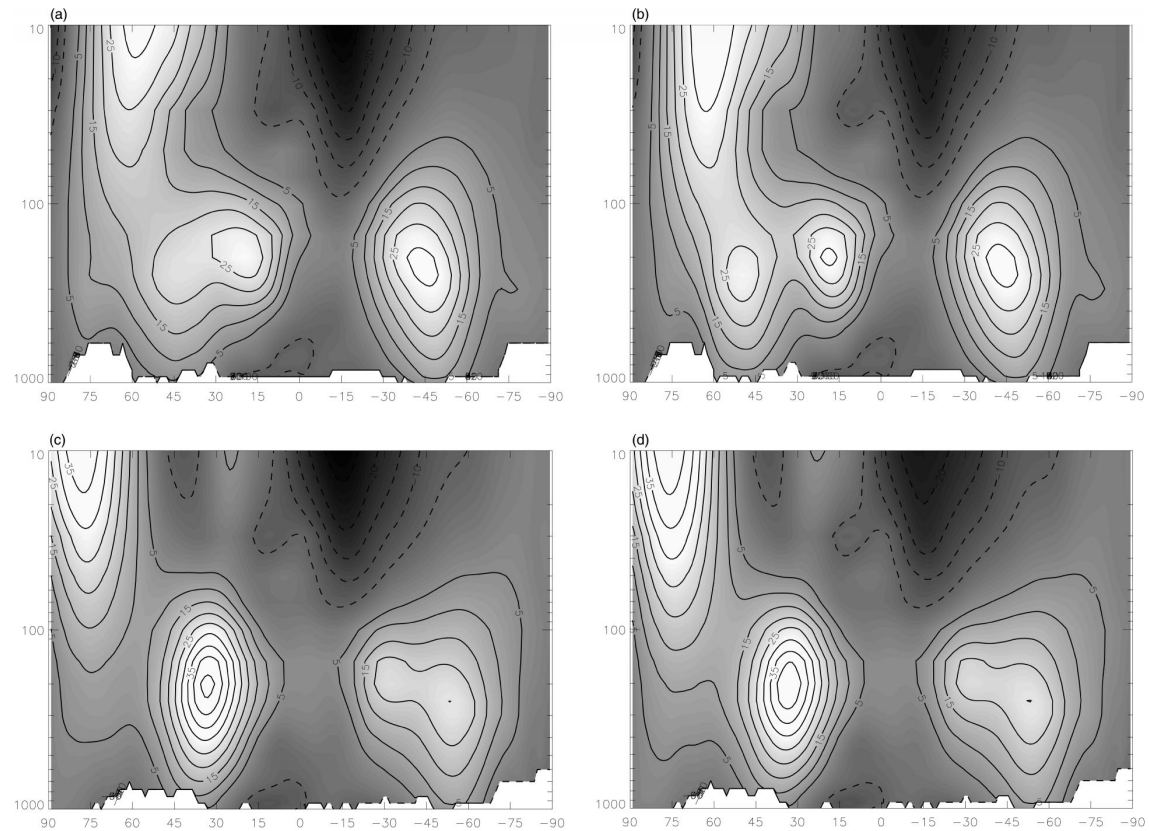


FIG. 6. Climatological mean of zonal winds in the Atlantic sector (60°W – 0°) for an AO index of (a) -1 std dev and (b) $+1$ std dev. (c) and (d) The same as (a) and (b), but for the Pacific sector zonal winds (150°E – 120°W).

Zonal wind in low (left) and high (right) AO index states, in Atlantic sector (top) and Pacific (bottom) (from Ambaum et al 2001)

Arguments

1. Annular Modes are important:
 - (a) EOFs show modes of hemispheric extent. The first EOF is almost zonally symmetric in Southern hemisphere.
 - (b) Dynamically:

Wave-meanflow interaction is the first mode of variability after linear baroclinic instability.

Turbulent flow in differentially rotating systems tends toward zonal structures (e.g. β -plane jets.)
2. Annular Modes are unlikely to be important:
 - (a) EOFs can be misleading — zonally symmetric EOFs do *not* indicate coherent zonally symmetric dynamics.
 - (b) Little evidence for statistically significant correlations of hemispheric extent; variance in (barotropic) zonally symmetric $m = 0$ modes is relatively small.
 - (c) Dynamically:

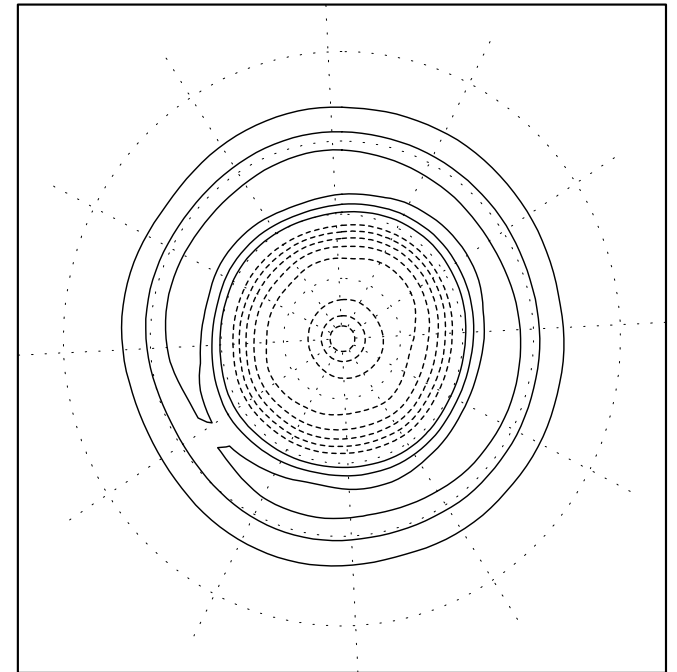
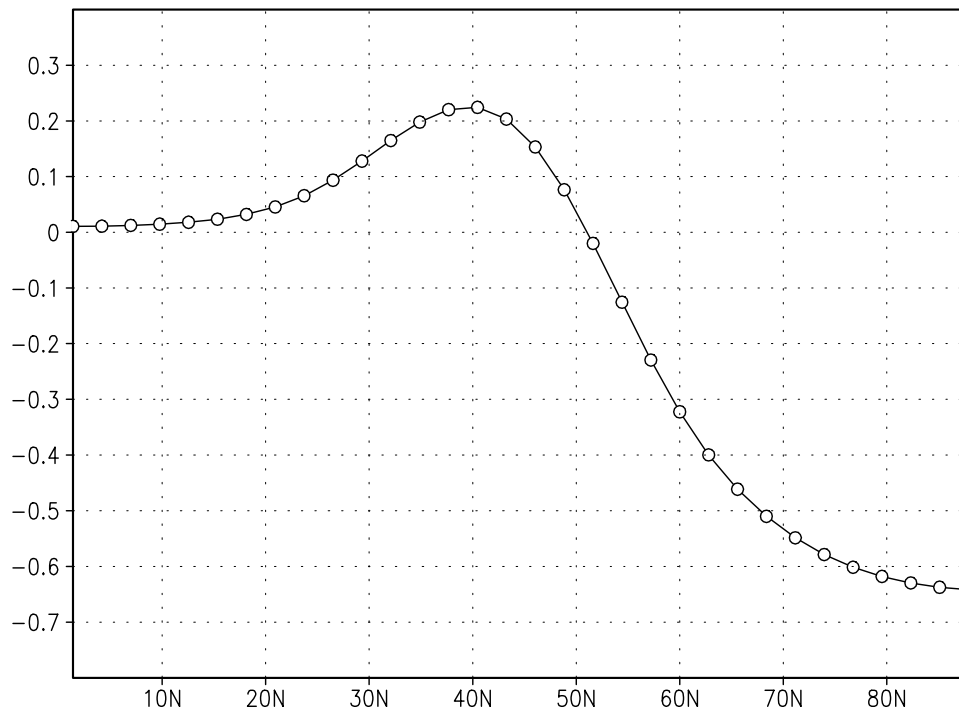
Atmosphere is not in a weakly nonlinear regime.

Inverse cascade (presuming its existence) is likely halted before significant energy reaches the zonally symmetric modes.

Models

1. A full GCM, with various idealized geometries.
T42, 14 levels, radiation + moisture + seasonal cycle.
Mixed layer ocean as lower boundary condition:
 - (a) Zonally symmetric (40m mixed layer)
 - (b) ZOnally assymmetric, either through topography ('Tibet') or variations in mixed layer thickness — land (20cm) and ocean (40m).
2. A balanced model. Barotropic vorticity equation, with stochastic forcing (at zonal wavenumbers 3 – 8, meridionally concentrated in mid-latitudes), representing baroclinic instability. Forcing is:
 - (a) Statistically zonally symmetric, or
 - (b) Statistically longitudinally concentrated in 'storm track' regions $\sim 60^\circ$ wide.

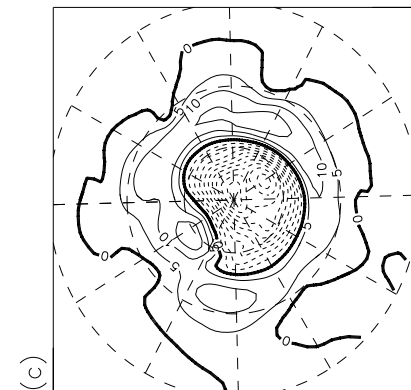
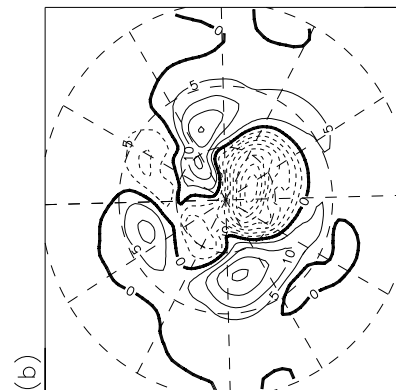
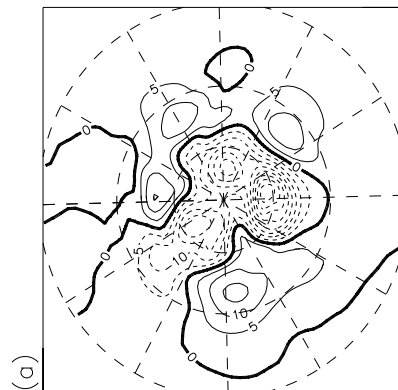
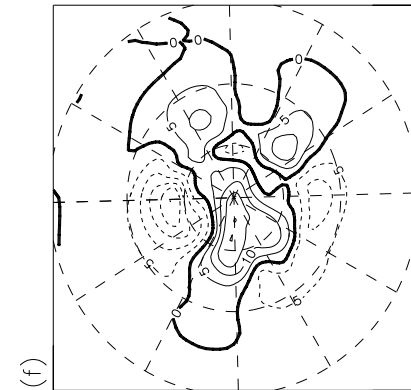
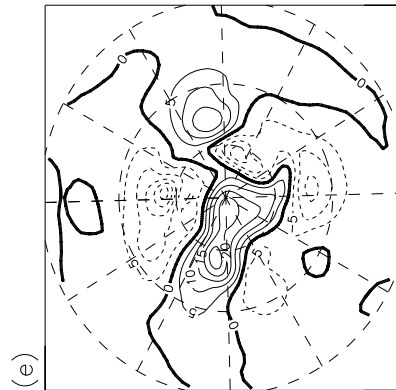
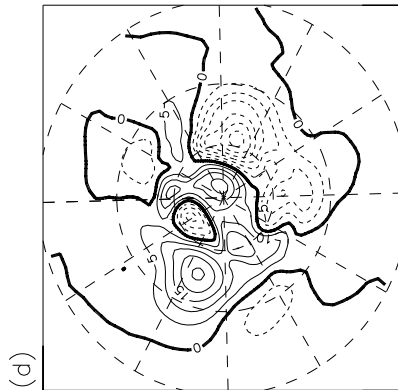
First EOF from zonally symmetric GCM.



The EOF itself is virtually zonally symmetric.

High and low index *events* (zonally symmetric model)

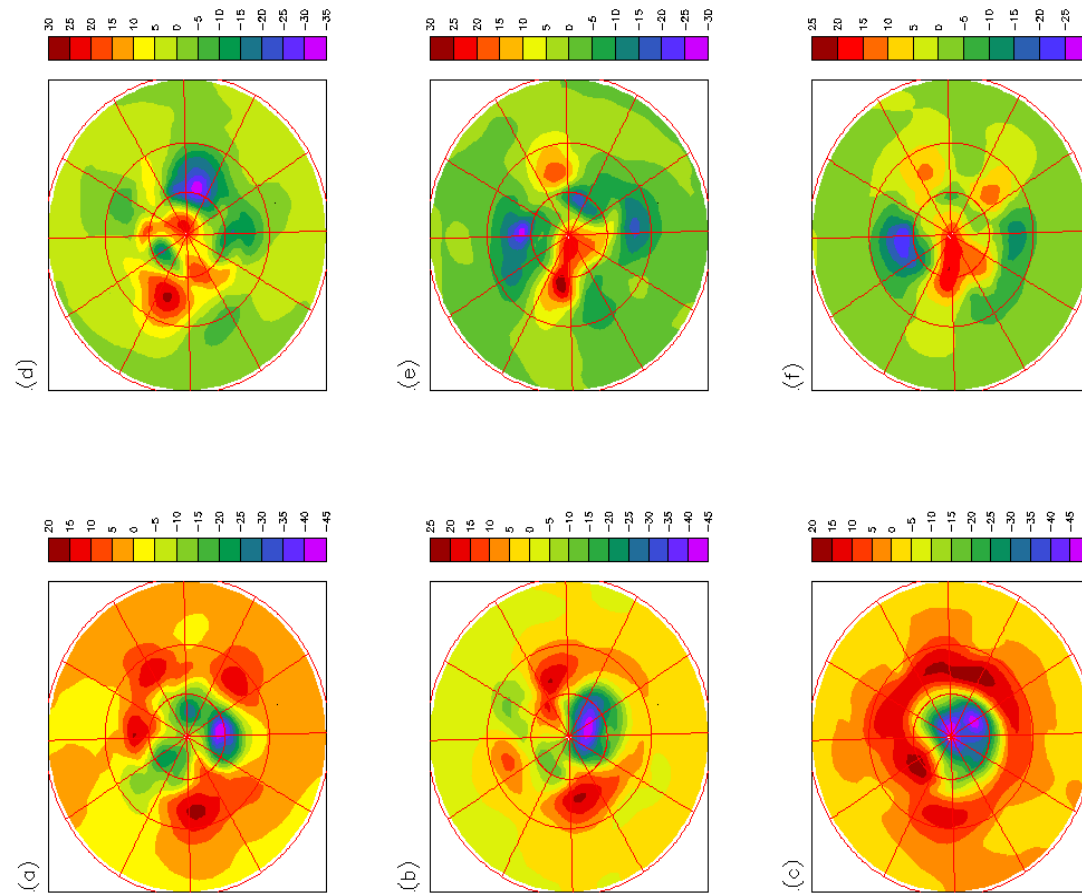
Low index:



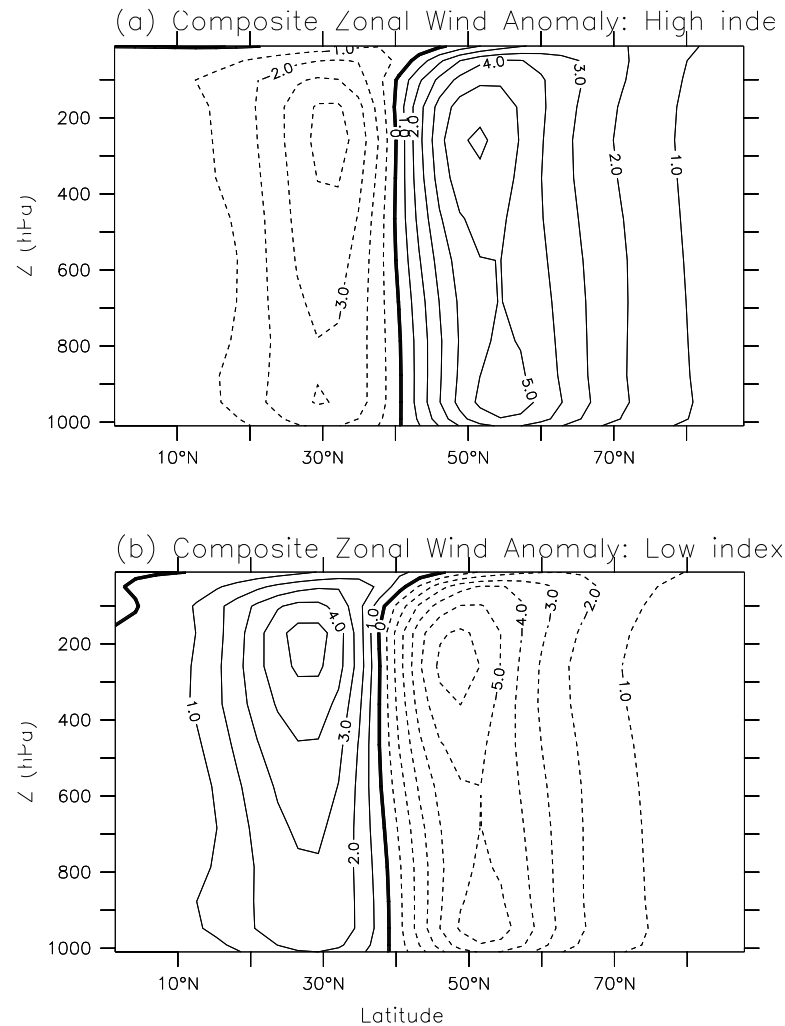
High index

High and low index *events* (zonally symmetric model)

Low index



High index

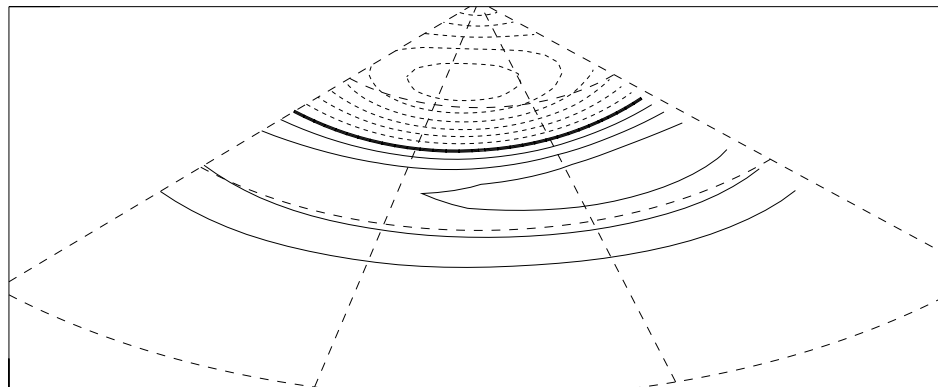


Composite zonal wind anomaly in GCM for high index (top) and low index (bottom) integration, with zonally symmetric boundary conditions.

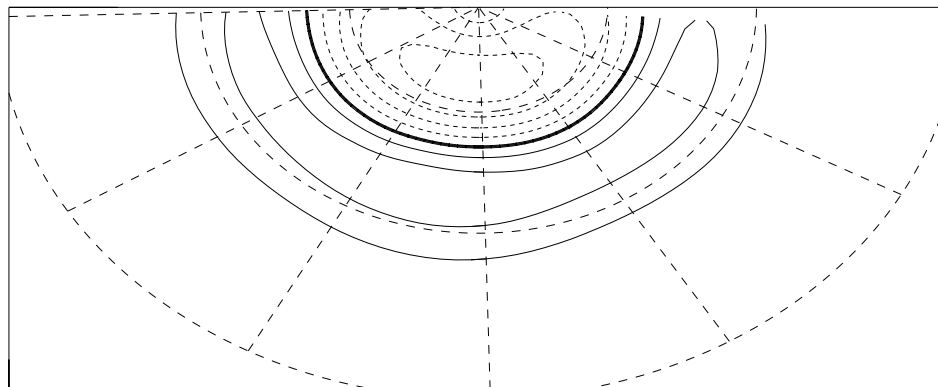
EOFs from a limited domain

$90^\circ \times 90^\circ$

(a)



(b)



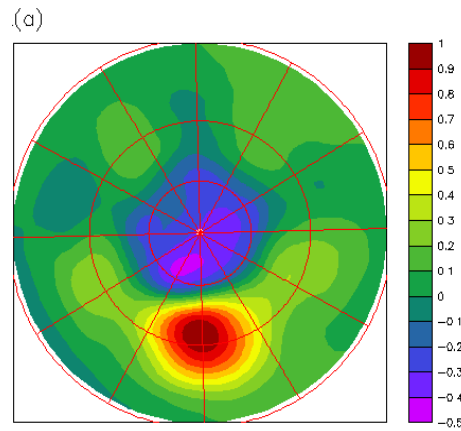
$180^\circ \times 90^\circ$

One-point correlation

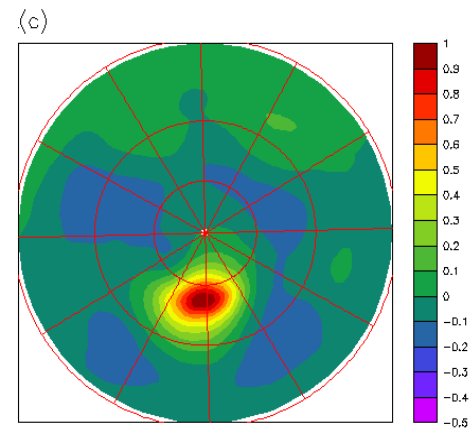
- ⑥ Correlation of a chosen point (the base point) with every other point in the domain. Related to energy spectrum.
- ⑥ May be simultaneous or lagged.
- ⑥ Provides a more direct measure of whether distant events are related.
- ⑥ Geographically separated regions of strong correlation/anti-correlation are 'teleconnections' (Namias, Wallace & Gutzler, 1981)

Correlations - (GCM)

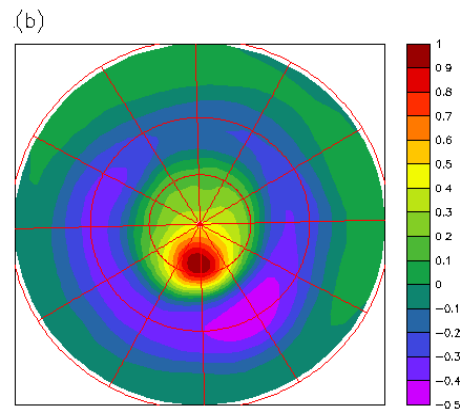
EOF maximum (lower)



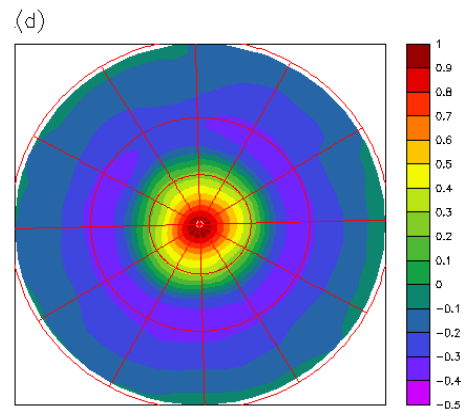
EOF minimum



OF maximim (upper)



Pole



Simple Theory: Generation of Zonal Motion

Zonal motion can be generated by random stirring on a β -plane, a crude parameterization of the effects of baroclinic instability on the barotropic mode.

$$\text{Zonal momentum equation: } \frac{\partial \bar{u}}{\partial t} = \overline{v' \zeta'} - \kappa \bar{u}$$

$$\text{Barotropic vorticity equation: } \frac{\partial \zeta'}{\partial t} = -\bar{u} \frac{\partial \zeta'}{\partial x} - v' \bar{q}_y + S_\zeta - D_\zeta$$

where S_ζ is stirring and D_ζ is dissipation. Multiply by $\zeta' / \bar{q}_y$ and integrate in x :

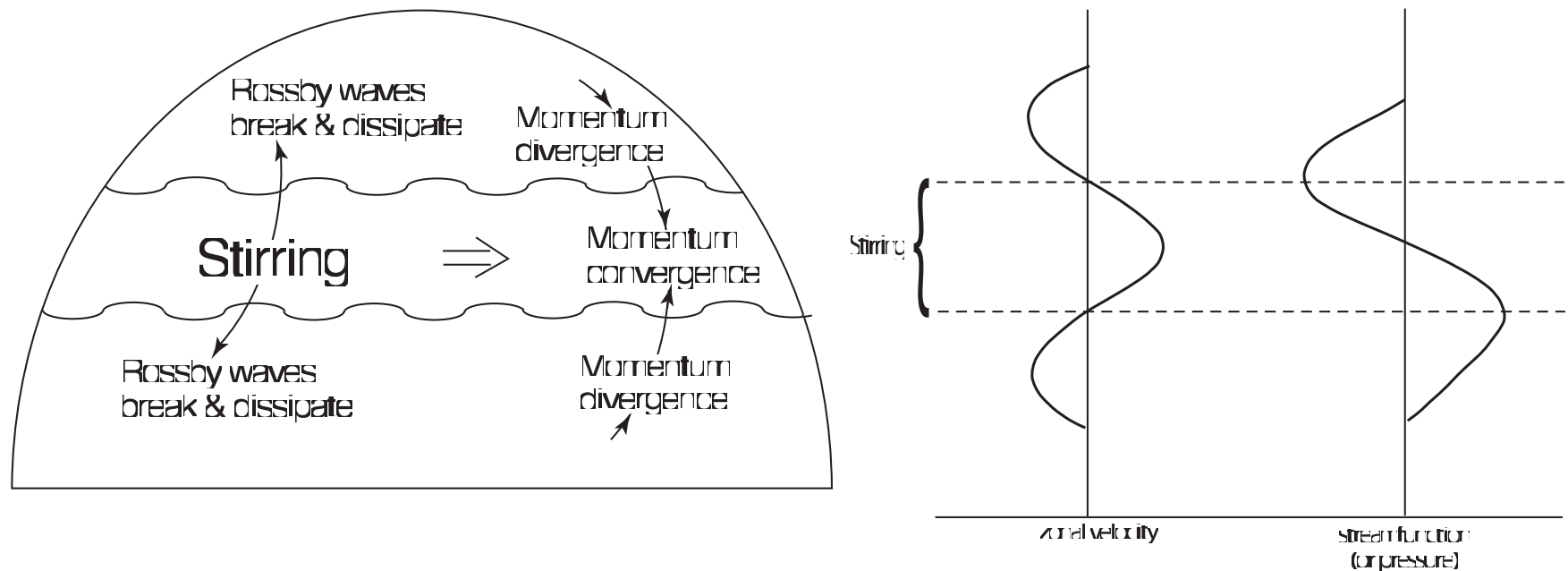
$$\text{Pseudo-momentum equation: } \frac{\partial P}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\overline{\zeta'^2}}{\bar{q}_y} \right) = -\overline{v' \zeta'} + S_P - D_P.$$

Statistically steady state:

$$\boxed{\kappa \bar{u} = S_P - D_P}$$

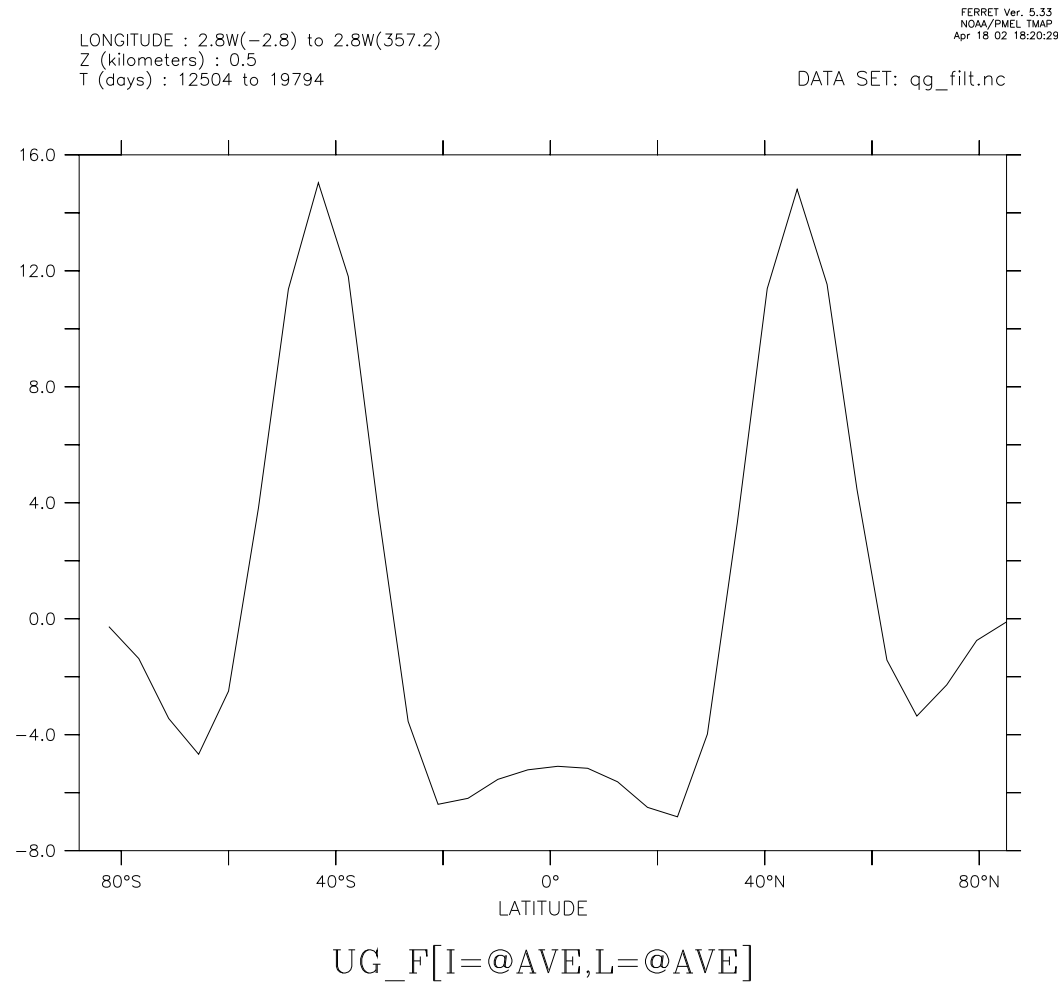
So westerlies are generated in regions of stirring (i.e. baroclinic instability) and these can be expected to vary in strength and in location.

Schematic of generation of eddy-driven jet



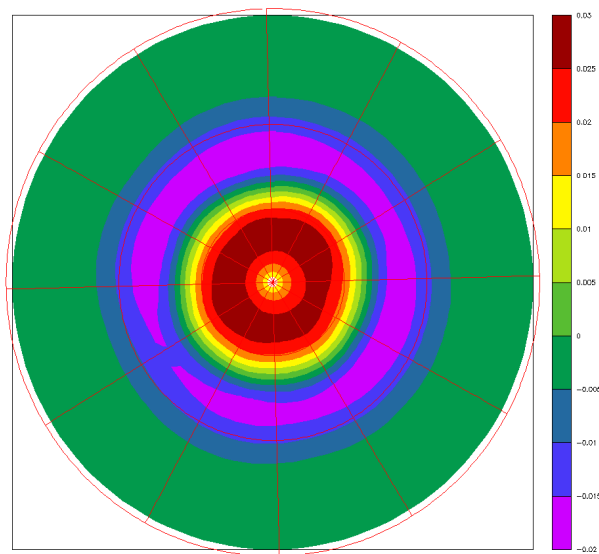
Stirring produces a *dipole* pattern of in the streamfunction (and the pressure).

Time & Zonally Averaged Flow in Barotropic model

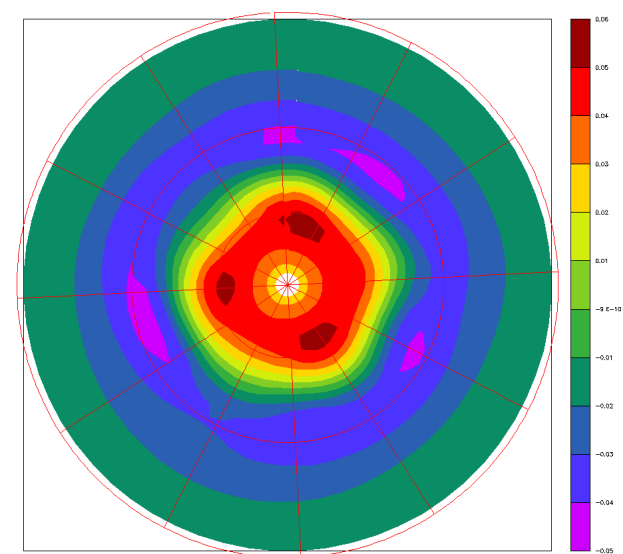


Leading EOF in statistically symmetric models

GCM

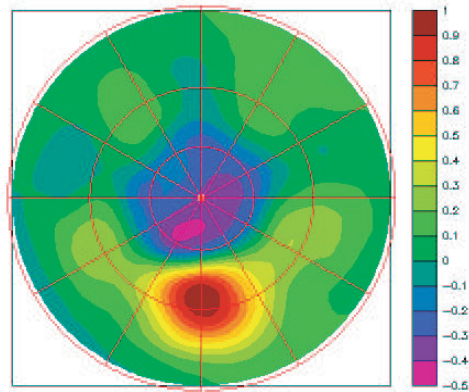


barotropic model

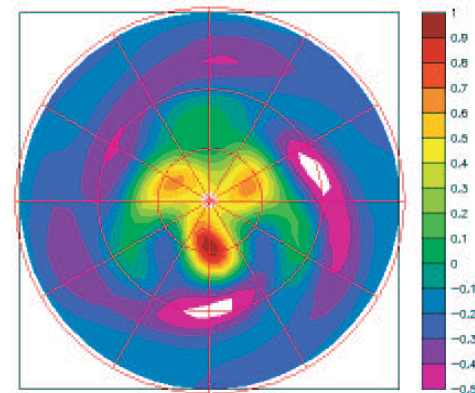
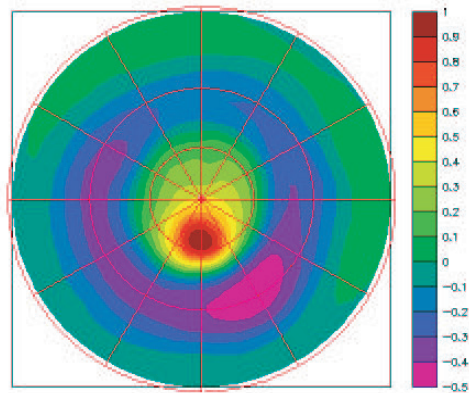
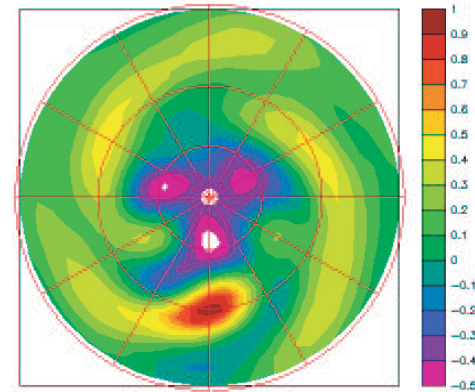


1 Pt Correlations, Zonally Symmetric MOdels:

GCM



Barotropic Model



Effects of Asymmetries

As we increase the asymmetry in the model, do the mechanisms of variability change qualitatively?

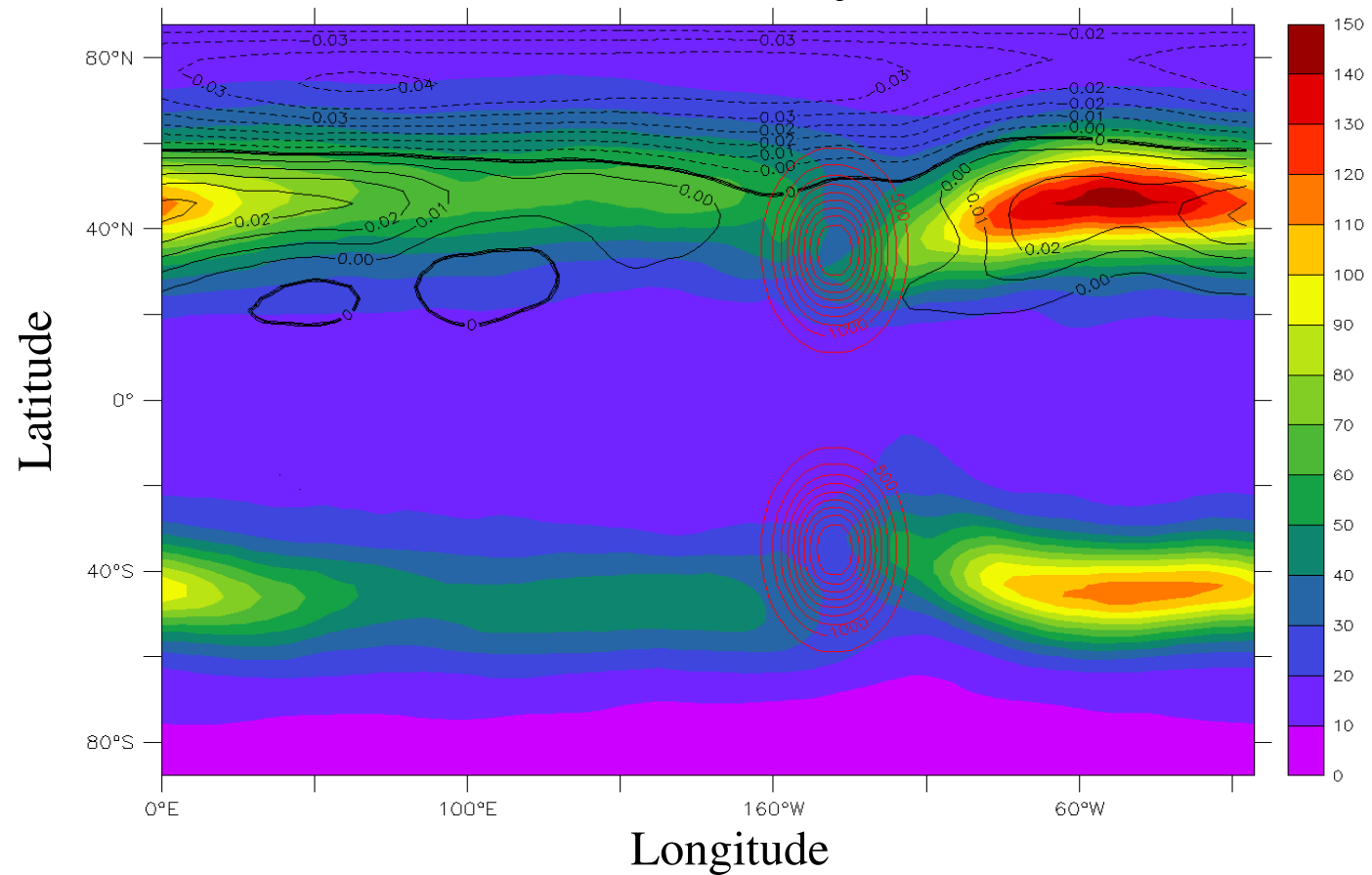
Is the EOF (a low frequency phenomena) tied to the location of eddy forcing by baroclinic instability (i.e. a high-frequency events)?

Use idealised mountains and land-sea contrasts in GCM.

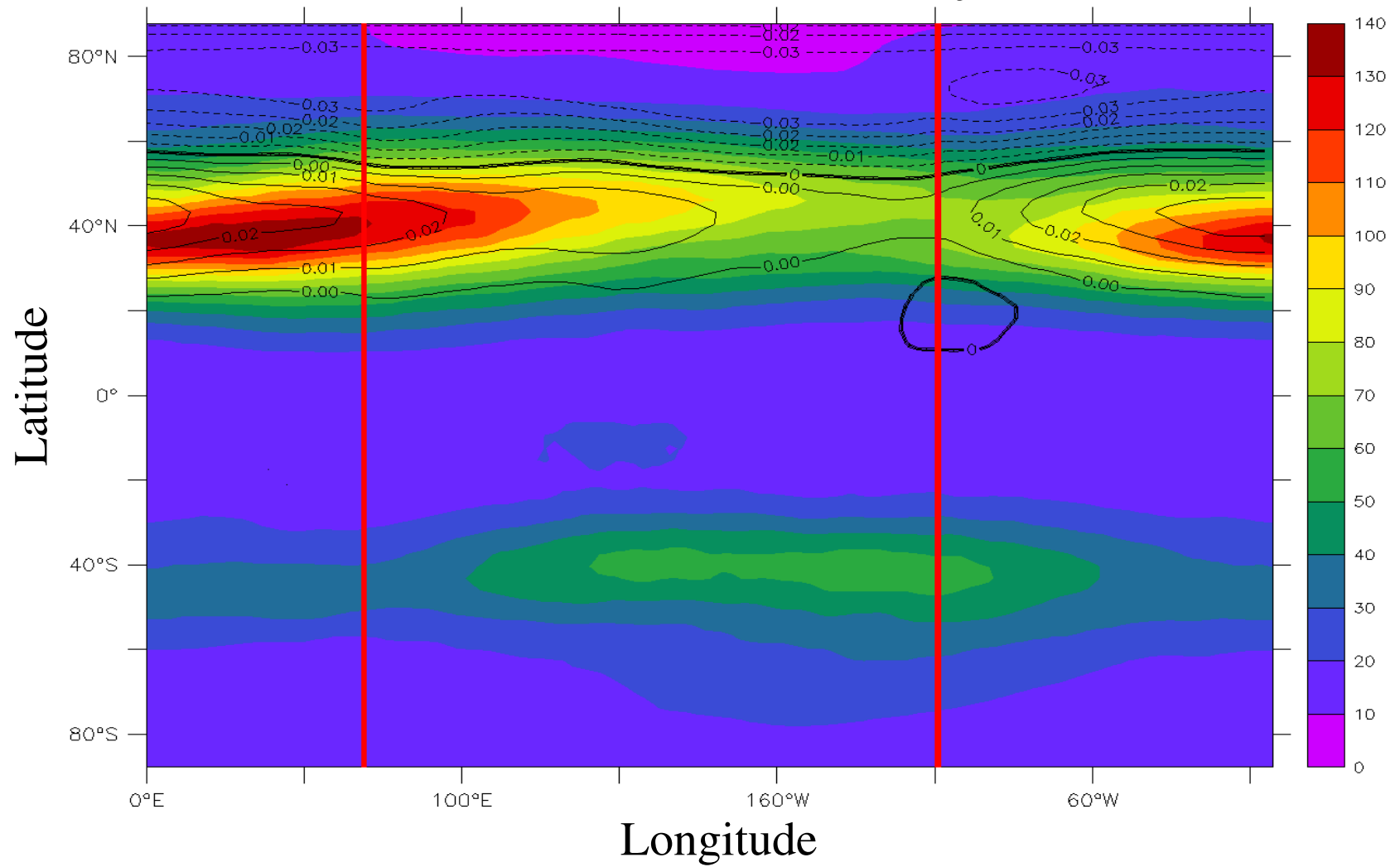
Vary intensity of stochastic forcing, with longitude, in barotropic model.

EOF (contours) and EKE (colours). GCM with 'Tibet'

Tibet Only

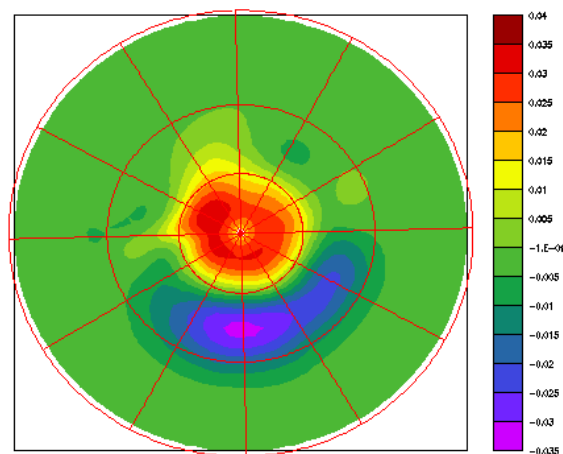


Land Sea Only

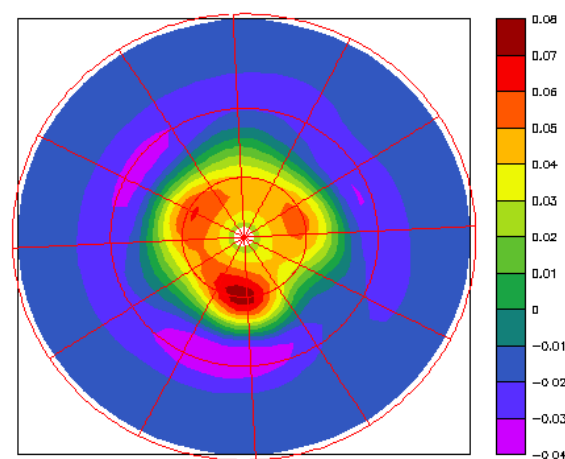


Leading EOF in asymmetric ('NAO-like') models

GCM

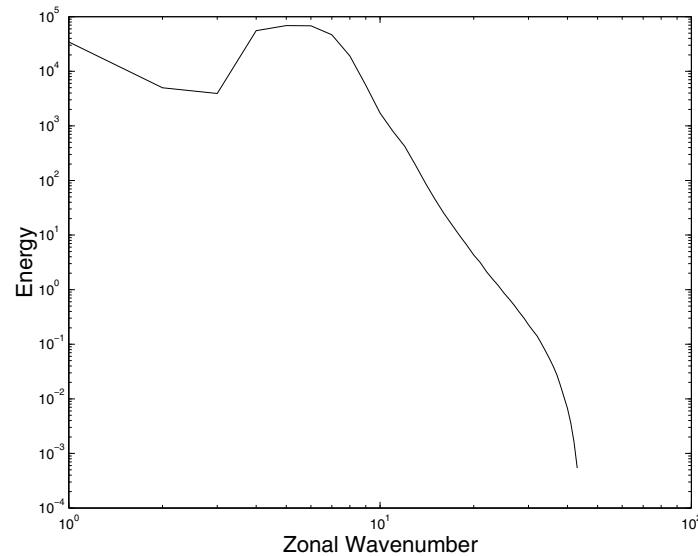


barotropic model

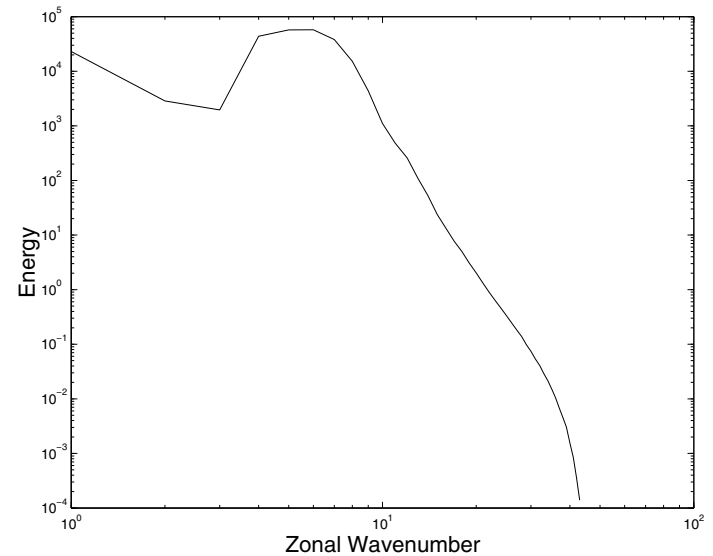


Spectra of Barotropic models

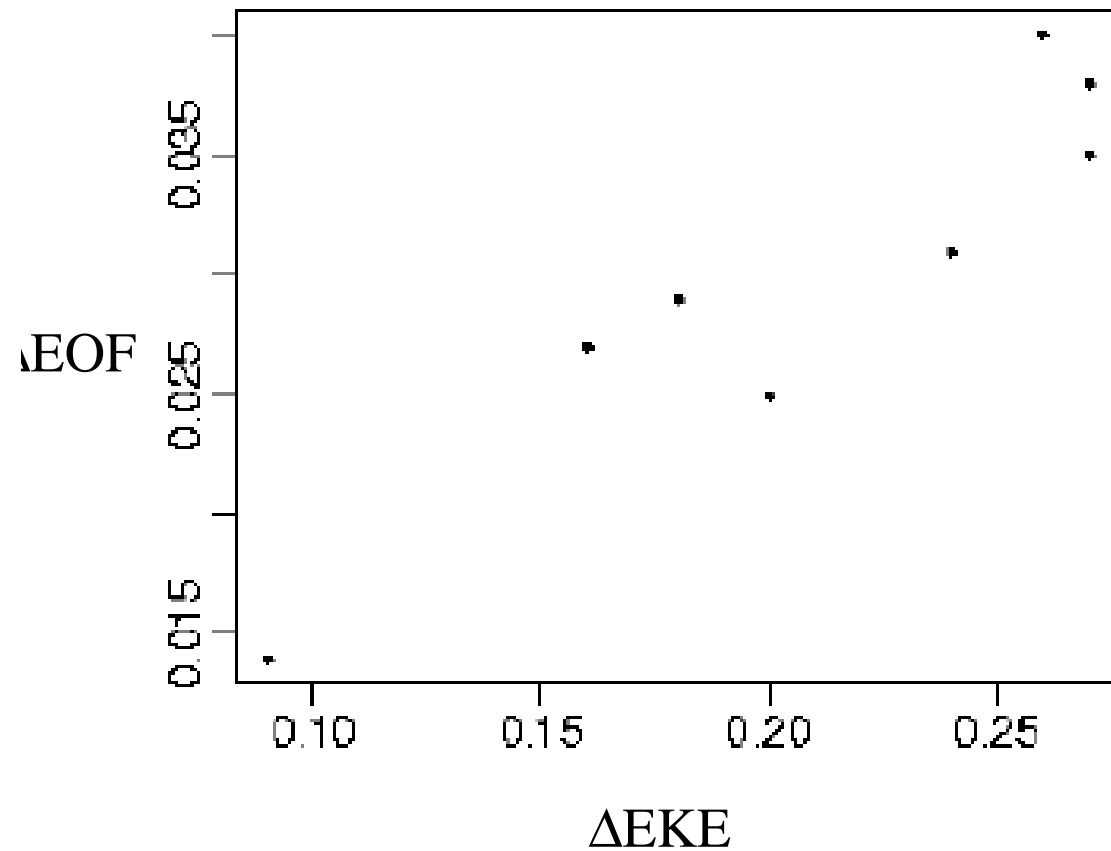
Zonally Symmetric



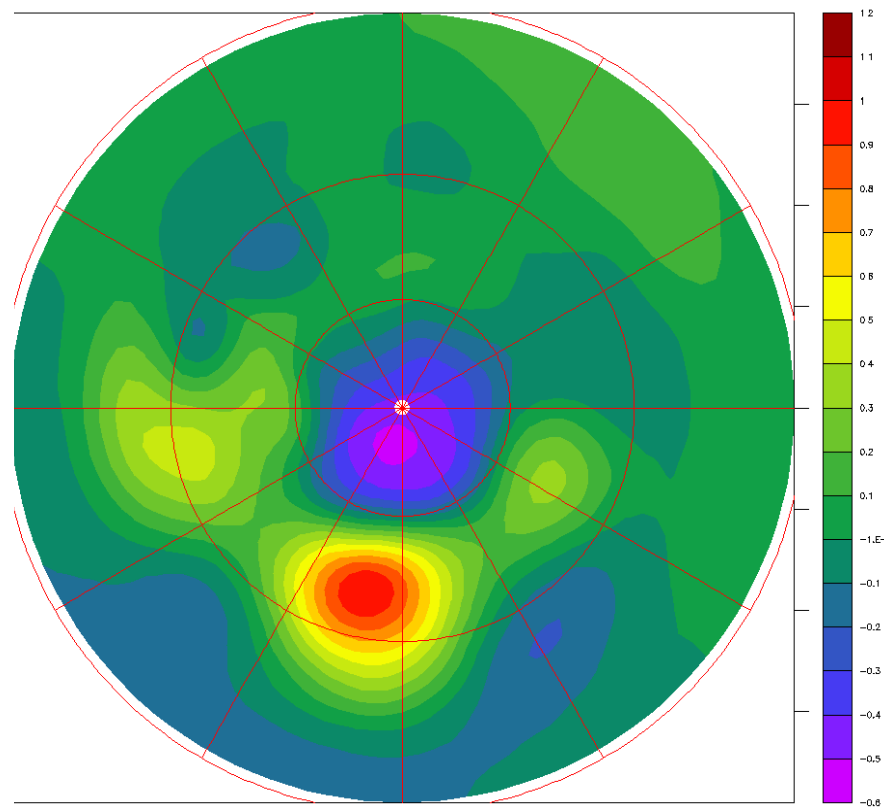
Zonally Asymmetric



Variance in EOF response vs EKE variance



Teleconnection Pattern Tibet Only Case



Conclusions

- (i) Dipole-like patterns are a natural consequence of forcing by meridionally localized baroclinic instability. Zonally localized forcing produces a corresponding localized response. NAO-like patterns are readily achievable, even in a barotropic, zonally symmetric model.
- (ii) Such forcing naturally produces eddy-momentum fluxes that in turn produce variability in the eddy-driven (subpolar, equivalent barotropic) jet. There may be oscillations in both the position and strength of the eddy-driven jet that contribute to the EOF.
- (iii) The annular mode, and the NAO, is the consequence of many zonally asymmetric 'events', of duration a few days (c.f. Feldstein). If the boundary conditions are zonally symmetric, it is averaging that produces an apparently zonally symmetric mode in the EOF.
- (iv) Neither the zonally symmetric GCM nor the zonally symmetric barotropic model produce 'annular modes' in a truly meaningful sense (i.e. dominant variability in $m = 0$ mode) when in a realistic parameter regime.
- (v) The timescale of the response (i.e. low-frequency variability) is produced by redenning of the eddy forcing (the input) by friction and nonlinear dynamics.